

ADITYA KUMAR SINGH

Jamshedpur, India

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Education

Kalinga Institute of Industrial Technology

B.Tech. in Computer Science, CGPA: 7.78/10

Sept 2022 – Present

Bhubaneswar, India

D.B.M.S English School (ISC)

Class XII, 84.5%

2022

D.B.M.S English School (ICSE)

Class X, 90.0%

2020

Research

Predicting Likelihood of Alzheimer's Disease using SVN

International Conference on Computational Intelligence (DoSCI-2025)

Mar 2025

- Developed an ML model using Support Vector Networks to assess Alzheimer's risk in elderly citizens.
- Presented research at DoSCI-2025 with collaboration from University of Delhi and Calabria, Italy.
- Focused on enhancing early detection accuracy in healthcare applications.

Projects

Autism Spectrum Disorder (ASD) Prediction | Python, Pandas, Random Forest

- Built a Random Forest model to predict ASD likelihood using behavioral and clinical data.
- Implemented preprocessing: missing values, categorical encoding, and feature scaling.
- Improved prediction reliability to support early diagnosis and decision-making.

Pothole Detection System | Python, OpenCV, YOLOv8, ByteTrack

- Developed a real-time pothole detection and tracking system achieving 91.4% mAP@50.
- Integrated GPS-based logging with CSV reporting for municipal maintenance workflows.
- Ensured robust detection across varied environments and road conditions.

Experience

Machine Learning Intern

Tata Steel Utilities and Infrastructure Services Ltd

May 2025 – July 2025

Jamshedpur

- Built a pothole detection system using YOLOv8 + ByteTrack with GPS-integrated reporting.
- Developed an RAG-based HR chatbot (LangChain, FAISS) with 94.2% query accuracy.
- Created Streamlit dashboards for monitoring and deployment of ML pipelines.
- Applied ML concepts to real-world industrial use cases in utilities and infrastructure.

Technical Skills

Programming Languages: Python, C/C++, Java, SQL

ML/AI: Scikit-learn, TensorFlow, PyTorch, YOLOv8, ByteTrack

NLP Tools: LangChain, HuggingFace, FAISS, RAG Pipelines

Data Science: Pandas, NumPy, Matplotlib, Seaborn, SciPy

Tools & Platforms: OpenCV, Streamlit, Flask, Git, Docker, VS Code, Jupyter

Certifications

Oracle AI Vector Search – Oracle University

OCI Data Science – Oracle University

Google Data Analytics – Coursera

Languages

English (Fluent)

Hindi (Native)